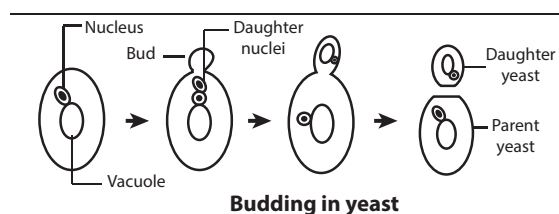


plants. As yeast grows, it converts its food through a process known as fermentation into alcohol and carbon dioxide. Fermentation is a part in industrial enzymes and chemicals, food products, industrial ethanol and malt beverage and wine, in diseases of humans, animals and plants, in food spoilage and as a model of molecular genetics. Those of the genus *Saccharomyces* ferment sugar and are used in the brewing and baking industries.



YEAST TWO-HYBRID SCREEN (TWO-HYBRID SCREENING; Y2H): A molecular biology technique for detecting protein-protein interactions and protein-DNA interactions in yeast cells. It can be used in high throughput screening of libraries of genes or potential genes from any species to identify pairs of genes that encode interacting proteins.

YELLOW CEREAL FLY: A pest which affects early-sown wheat crops and causes the death of the plant's central shoot. The larvae enter through the top of the shoot without leaving an entry hole.

YELLOW DWARF VIRUS (BYDV): A fungal disease affecting barley and also wheat and grass. The leaves turn red and yellow and yields are reduced. The disease is spread by wingless aphids.

YELLOW FEVER: An infectious viral disease of tropical origin transmitted by mosquitoes and characterised by high fever, haemorrhaging, vomiting of blood, liver damage and jaundice. The agent is a flavivirus, an arbovirus of group B. The virus multiplies in mosquitoes, which remain infectious for life.

YELLOW RATTLE: An annual weed, *Rhinanthus minor*, in the genus *Rhinanthus*, family Orobanchaceae. It is found in grasslands and native to Europe and western Asia.

YELLOW RUST (STRIPE RUST): A rust disease, *Puccinia striiformis*, of cereals, mainly affecting wheat and barley. Yellow pustules form on leaves, stems and ears.

YELLOW SALT (URANIUM NITRATE, URANYL NITRATE): A toxic, explosive, unstable yellow crystal with the chemical formula $\text{UO}_2(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, which melts at 60°C (333 K) and boils at 118°C (391 K). Yellow salt is soluble in water, alcohol and ether and used in medicine, photography, uranium extraction, etc.

YELLOW SPOT (FOVEA; MACULA RETINAE; SOEMMERING'S SPOT): The most sensitive part of the retina in the human eye at which visual perception is most acute. It is a yellowish area opposite the lens and contains a large number of cone cells.

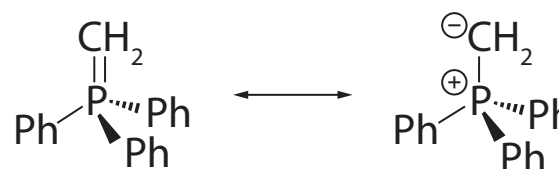
YELLOW-GREEN ALGAE (XANTHOPHYTA): A division consisting mainly of freshwater and terrestrial algae, including unicellular, colonial (palmelloid), filamentous and siphonaceous forms. They are characterised by having disclike yellow-green plastids (due to the large amounts of carotenoids present) and by accumulating oil and leucosin rather than starch. The motile stages have two unequal flagella (one flimmer and one smooth flagellum). The cell wall is divided into two halves but this character is not obvious except in the filamentous forms, where disruption results in a number of H-shaped fragments.

YELLOW: A general term for any of various fungus diseases of plants such as cabbage, lettuce, cauliflower, peach, sugar beet and other plants in which the leaves become yellow.

YIELD: 1. **Chemical yield** or **reaction yield** is the efficiency of a chemical reaction in changing reactants into products or the amount of products obtained in a chemical reaction. For example, a 50% yield means that half the reactants are converted into products. The

remainder, which is unchanged becomes by-products. The theoretical or ideal yield is therefore calculated based on the molar amount of the limiting reactant, taking into account the stoichiometry of the reaction. For the calculation, it is usually assumed that there is only one reaction involved. The fractional yield, relative yield or percentage yield, which serve to measure the effectiveness of a synthetic procedure, is calculated by dividing the amount of the obtained product in moles by the theoretical yield in moles: fractional yield = actual yield/theoretical yield. 2. **Yield strength** or **yield point** in a material is the stress at which a material begins to deform plastically after first deforming elastically at which point it can return to its original shape when the applied stress is removed. Once the yield point is passed some fraction of the deformation will be permanent and non-reversible. Knowledge of the yield point is vital when designing a component since it generally represents an upper limit to the load that can be applied.

YLIDE: A neutral molecule having negative charges such as carbon, nitrogen, oxygen or sulfur adjacent to a positive heteroatom such as phosphorus, nitrogen, sulfur, selenium and oxygen or an organic compound which contains two adjacent atoms bearing formal positive and negative charges and in which both atoms have full octets of electrons. Ylides are very reactive and are useful in organic synthesis such as Wittig reaction.



Chemical Structure of Ylide

YOCTO-: Prefix used in the metric system to denote 10^{-24} . Yocto is the smallest confirmed SI prefix. Its symbol is y. For example, 1 yoctamine (ym) = 10^{-24} minute.

YOGHURT: Semi-solid milk fermented by bacteria to give a slightly sour flavour. It is made traditionally by adding common strains of *Streptococcus* and *Lactobacillus* bacteria to raw milk. Commercially, a blend of concentrated sterilised milk and milk solids is inoculated with the two bacteria, sometimes *L. acidophilus* or lactose-fermenting yeast is also added. The product is then incubated four or five hours at 44°C (317 K) until curd forms.

YOKE: 1. A device attached to the neck of a cathode ray tube to control the scanning motion of the electron beam. 2. A wooden frame with a double harness to connect the collar of each horse to the pole of a wagon or coach. 3. A frame designed to be carried across a person's shoulders with equal loads suspended from each end. 4. A series of two or more magnetic recording heads fastened securely together for playing or recording on more than one track simultaneously. 5. A clamp or vise that holds a machine part in place or controls its movement or that holds two such parts together. 6. The handle of the steering mechanism for an airplane's ailerons.

YOLK: 1. The food stored in an egg, which is yellow, usually the spherical portion of an egg of a bird or reptile consisting mostly of proteins and fats, surrounded by the albumen and used by the growing embryo. The largest yolks are contained in bird, reptile and fish eggs. 2. A greasy substance found in unprocessed sheep's wool.

YOLK CORD: A slender protoplasmic cord that connects the yolk glands with the egg chambers in an insect such as an aphid.

YOLK PLATELETS (YOLK BODY): A membrane-bound storage structure, containing protein and lipid that is found in large numbers in the cytoplasm of the eggs of all animals except mammals. They are the most prominent component in the cytoplasm of amphibian eggs and embryonic cells and have been generally designated as inert reservoirs of raw material to be utilised during subsequent embryonic and larval development.

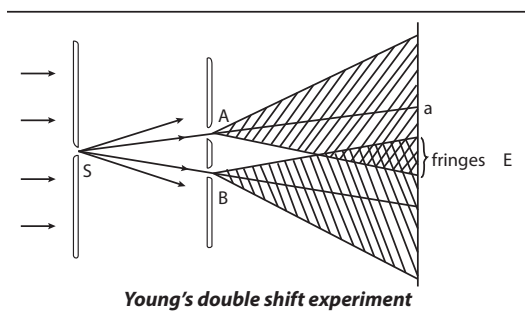
YOLK PLUG: The remaining patch of endodermal cells that is created during the formation of the ventral lip of the blastopore. It is a patch of large endodermal cells which remains exposed on the vegetal surface of the amphibian blastula that will eventually be internalised by epiboly.

YOLK SAC (VITELLINE SAC): A membranous sac attached to an embryo, providing early nourishment in the form of yolk in bony fishes, sharks, reptiles, birds and primitive mammals and functioning as the circulatory system of the human embryo before internal circulation begins.

YOTTA: A prefix used in the metric system to denote 10^{24} . It is the largest unit prefix in the International System of Units (SI). Its symbol is Y. For example, 10^{24} joules = 1 yottajoule (Yj).

YOUNG, THOMAS (1773-1829): British physician and physicist, who was a child prodigy and spoke fourteen languages before he was nineteen years old. He trained as a physician and practised medicine at St. George's Hospital from 1811 until his death. He also spent much of his time on scientific researches. His early researches concerned the eye and vision, but he is best known for establishing the wave theory of light (1800-1804) and explaining the phenomenon of interference (1807). The Young modulus of elasticity is named after him.

YOUNG'S DOUBLE-SLIT EXPERIMENT (YOUNG'S INTERFERENCE EXPERIMENT; YOUNG'S DOUBLE-SLIT INTERFEROMETER; DOUBLE-SLIT EXPERIMENT; YOUNG'S EXPERIMENT): A nineteenth-century investigation into the properties of light that has since been found to demonstrate both the duality (wave and particulate nature) of photons and the concepts of superposition and quantum interference. It involves particle beams or coherent waves passing through two closely-spaced slits, after which in many circumstances they are found to interfere with each other. It verifies that light is a wave simply because of the bright and dark fringes that appear on a screen. It is the constructive and destructive interference of light waves that cause such fringes.



YOUNG'S MODULUS (YOUNG MODULUS OF ELASTICITY; TENSILE MODULUS): A measure of the stiffness of an isotropic elastic material, defined as the ratio of the uniaxial stress over the uniaxial strain in the range of stress in which Hooke's Law holds. According to Hooke's law, the strain is proportional to stress and therefore, the ratio of the two is a constant that is commonly used to indicate the elasticity of the substance. It can be experimentally determined from the slope of a stress-strain curve created during tensile tests conducted on a sample of the material. It is expressed

as: $Y = \frac{\text{stress}}{\text{strain}}$. It is named after after Thomas Young (1773-1829),

an English physician and physicist.

YOUTUBE: A video sharing service that allows users to watch videos posted by other users and upload videos of their own. The service was started as an independent website in 2005 and was acquired by Google in 2006. Videos that have been uploaded to YouTube may appear on the YouTube website and can also be posted on other websites, though the files are hosted on the YouTube server.

YTTERBIUM: A silvery-white metallic element in Group 3 of the Periodic Table (rare-earth group including the lanthanides) with the chemical symbol Yb, atomic number 70 and atomic weight 173.04, melting point 819 °C (1092 K), boiling point 1196 °C (1469 K) and a relative density of 7. It has seven natural isotopes and ten artificial isotopes are known. The common oxide, Yb_2O_3 , is colourless and dissolves readily in acids to form colourless solutions of trivalent salts which are paramagnetic. Ytterbium also forms a series of divalent compounds. The divalent salts are soluble in water but react very slowly with water to liberate hydrogen. It is used in certain steels.

YTTRIUM: A grey metallic element in Group 3 of the Periodic Table (rare-earth group including the lanthanides) with the chemical symbol Y, atomic number 39 and atomic weight 88.905, melting point 1522 °C (1795 K), boils at about 3345 °C (3618 K) and a relative density of 4.47. The metal is stable in air below 400 °C (673 K). It occurs in uranium and lanthanoid ores from which it can be obtained by an ion exchange process. Yttrium metal absorbs hydrogen and in alloys up to a composition of YH_2 they resemble metals very closely. In certain composition ranges, the alloy is a better conductor of electricity than the pure metal. It is used in alloys for super conductors and magnets. Yttrium (VI) oxide, Y_2O_6 , is used in lasers and phosphors. Yttrium acetate, yttrium chloride and yttrium sulfate are all used as analytical reagents. The natural stable isotope is yttrium-89 (^{89}Y) and there are fourteen known artificial isotopes.

ADVERT SPACE

Contact

+233 (0)24 891 9104



Z: 1. The symbol for the atomic number of an element, which is the number of protons in the atom. 2. The symbol for impedance, which is a measure of the total opposition to current flow in an alternating current circuit. It is made up of two components, ohmic resistance and reactance and usually represented in complex notation as $Z = R + iX$, where R is the ohmic resistance and X is the reactance. 3. Written as $\mathbb{Z}$ or $\mathbb{I}$, it is the symbol for the set of integers. Written as $\mathbb{Z}^+$, it is the symbol for the set of positive integers. Written as $\mathbb{Z}_n$, it is the symbol for the ring of modular arithmetic modulo n .

Z BOSON (Z PARTICLE): An electrically neutral elementary particle with zero electrical (z^0) charge. It is its own antiparticle that acts upon all known subatomic particles and the neutral partner of the electrically charged W particle. It has a mass of about 90 GeV, about 182,000 times that of the electron and about 100 times more massive than the proton. It has a lifetime of only about 10^{-25} second. It constitutes the quantum of weak interactions in which the charges of participating particles do not change.

Z FORM: One of the three biologically active double helical structures together with A- and B-DNA (B- DNA predominates in the cell). It is a double stranded DNA containing strands of alternating purines and pyrimidines, in which the helix formed by the two polynucleotide chains is left-handed rather than right-handed, forming a zigzag look of this DNA backbone when viewed from the side. Z-DNA may be involved in regulating given expression in eukaryotes.

Z LINE (Z BAND; Z-MEMBRANE, ZWISCHEN-LINE): The dark thin protein band to which actin filaments are attached that separates adjacent sarcomeres in the fibres of striated muscle. It is visible under electron microscope as a thin dark line in the middle of a lighter band. The centre point of each thin filament is anchored to the Z line and exert a pull on the membrane when the fibre contracts.

Z SCORE (STANDARD SCORE): A statistical measurement, that expresses the degree of standard deviations from the mean of the distribution scores in a group; it is the normal random variable of a standard normal distribution. A z-score of 0 means the score is the same as the mean; if less than 0, the score is less than the mean; if greater than 0, the score is greater than the mean; if equal to 1 the score is 1 standard deviation greater than the mean; if equal to 2, the score is 2 standard deviations greater than the mean; etc.

Z-AXIS: The third dimension in three-dimensional Cartesian coordinate system, which is vertical to the ground and directed upward and downward, with upward being designated positive and downward, negative.

Z-BUFFER: In computer science, it is a buffer for storing depth information for displaying three-dimensional graphics, which

implies displaying graphics using x , y and z coordinates. Two-dimensional images are displayed using x , y coordinates. In a graphics card, z-buffer memory keeps track of which onscreen elements are visible and which are hidden behind other objects.

Z-COORDINATE: The position of a point along the z -axis in a three-dimensional Cartesian coordinate system, which is equal to the directed distance of a point.

Z-DISTRIBUTION (STANDARD NORMAL DISTRIBUTION):

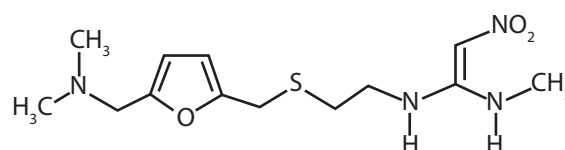
In statistics, a special case of normal distribution when a normal random variable has a mean of 0 and a standard deviation of 1. The normal random variable of a standard normal distribution is called a **standard score** or a **z score**, which is the result of converting from normal distribution to standard normal distribution by the formula $z = (X - \mu)/\sigma$, where X is a normal random variable, μ is the mean and σ is the standard deviation.

Z-TEST (NORMAL TEST): A test using a sample of n independent observations of the null hypothesis that a normal distribution with a variance σ^2 has mean μ , with test statistics z , given by $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \cdot \bar{x}$

is the sample mean, if $|z| > 1.96$ is evidence for the rejection of the null hypothesis of the 5% significance level in favour of the alternative which says that the mean is not μ .

Z-VALUE: A value, usually of a test statistic that is hypothesised to have a normal distribution with mean 0 and variance of 1. This z -notation was introduced by Sir Ronald Fischer in 1924.

ZANTAC (ZINETAC; RANITIDINE; HYDROCHLORIDE): The name for an anti-ulcer drug with the chemical formula $C_{13}H_{22}N_4O_3S$. It is a histamine H_2 -receptor antagonist that inhibits stomach acid production. It is used to treat peptic ulcers and relieve heart burns.



Chemical Structure of Zantac

ZARISKI TOPOLOGY: In mathematics, it is a topology on an infinite set whose open sets are those with finite complements; it is not metrizable as it is not Hausdorff. It is named after Oscar Zariski (1899 - 1986), Russian-born American mathematician.

ZEa: A genus of true grasses in the family Poaceae. Several species are commonly known as teosintes and are found in Mexico, Guatemala and Nicaragua. There are five recognised species in the genus, namely *Zea diploperennis*, *Zea perennis*, *Zea luxurians*, *Zea nicaraguensis* and *Zea mays*.

ZEATIN: A naturally occurring cytokinin derived from the purine adenine, present in immature grains of the maize plant (*Zea mays*). Zeatin and derivatives were discovered to be the primary active ingredient in coconut milk, which has been known to actively induce plant growth.

ZEBROID (ZEBRA MULE; ZEBRULE): A hybrid animal that results from the cross between a zebra and any other equine or horse-like animal. The zebroid term is a very general one, referring to various hybrid animals. In most cases the male is a zebra, with the female being a horse, a donkey, etc.

ZEBU (BRAHMAN): A humped domestic cattle, *Bos indicus*, indigenous to India, where it is sacred and used as a draft animal as well as for producing milk. It has a pronounced shoulder hump, long